

# WASP-12b

R-0548

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_180222 · created 2026-07-22T07:16:43+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458172.87347 +/- 0.00048 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.05 +/- 0.044                    |
| Transit depth (Rp/Rs)^2                           | 0.25 +/- 0.44 [%]                 |
| Orbital Inclination (inc)                         | 86.46 +/- 2.91                    |
| Airmass coefficient 1 (a1)                        | 0.87 +/- 0.43                     |
| Airmass coefficient 2 (a2)                        | 0.052 +/- 0.08                    |
| Scatter in the residuals of the lightcurve fit is | 48.4 %                            |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 2.39                              |
| Optimal Annulus                                   | 7.25                              |
| Transit Duration (day)                            | 0.1139 +/- 0.0032                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.05 ± 0.044 vs 0.1178 (deviation 1.54σ) |
| Duration fit vs published | 0.1139 d vs 0.125 d (deviation 3.47σ)    |
| Mid-time O–C              | 1572.3 min                               |
| Depth SNR                 | 1.1                                      |
| Residual scatter          | 48.4 %                                   |

### Light curve

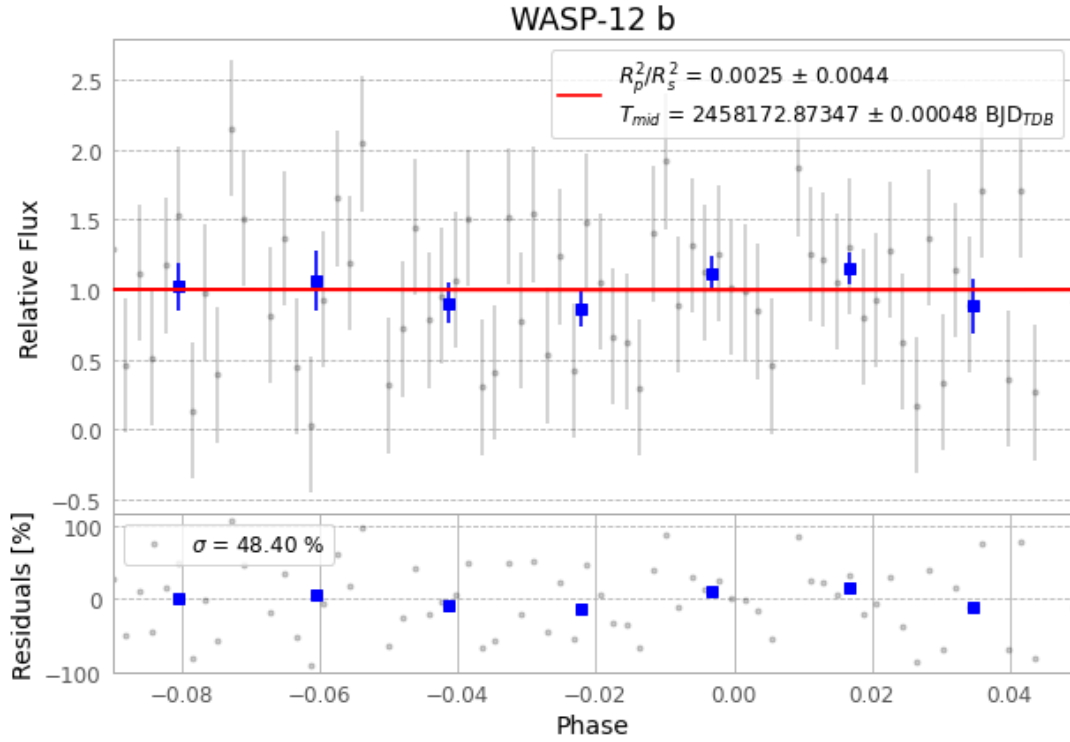


Figure 1 — FinalLightCurve\_WASP-12 b\_2018-02-22.png (SHA-256 facdd976a1f39ebc...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [632, 466], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 938244f0091ff2a9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

74 FITS frames (49 MB), WASP-12180223063012.FITS → WASP-12180223100912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-180222.log (70c2927bb27b...), FinalLightCurve\_WASP-12 b\_2018-02-22.png (facdd976a1f3...), FinalLightCurve\_WASP-12 b\_2018-02-22.csv (68da8cee059c...)

## Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 07:16Z.